

Noether Constraints and the Measure of Compatible Histories

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Abstract

The accumulated-history approach to cosmology begins with a striking scale connection: if a three-dimensional state acquires three effective distinctions per doubling of cosmic age, then the Hubble expansion rate is $H = 1/t$, and the benchmark age of the Universe gives $H_0 = 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The next question is more fundamental than the numerical coincidence: what makes two histories compatible, and what measure turns their disappearance into information? This note develops that missing layer through a sequence of controlled tests. A conditional survival estimator correctly removes predictable expansion and deterministic chaos while recovering injected information rates. A geometric tolerance, however, does not define a unique continuum bit rate. Conservation then supplies physical admissibility. Following Noether's connection between time-translation symmetry and energy conservation, candidate transitions are restricted to cells intersecting an energy shell, and accumulated paths are restricted by a discrete stationary-action equation. The raw cardinalities exhibit universal spatial and temporal refinement terms. Their finite cell-count remainder converges but is not invariant under grid deformation or canonical coordinates. Liouville phase-volume density is different: it agrees across those choices to better than 0.26% for both harmonic and anharmonic Hamiltonians. The result advances the program from an arbitrary neighborhood count to an invariant measure and isolates the remaining physical task: define a covariant conditional history measure and its continuum time normalization.

From the Hubble scale to a measurement problem

The preceding note, *Hubble Scale from Time-Induced Complexity*, began with a simple relational theorem. When nonnegative distinguishing coordinates are appended to an additive distance, fixed-radius neighborhoods can only lose members. If $\rho(t)$ denotes the fraction of alternative histories still compatible with a present history, define

$$C(t) \equiv -\log_2 \rho(t). \quad (1)$$

Here t is cosmic time and $\log_2$ is the base-two logarithm, so C is measured in bits. Identifying inverse compatible-history density with a dimensionless relational volume and taking its three-dimensional linear scale gives the proposed bridge

$$a_{\text{rel}}(t) \propto 2^{C(t)/3}, \quad H_{\text{rel}}(t) = \frac{\ln 2}{3} \dot{C}(t). \quad (2)$$

Here a_{rel} is the hypothesized relational scale factor, $H_{\text{rel}} \equiv \dot{a}_{\text{rel}}/a_{\text{rel}}$ is its fractional expansion rate, an overdot means d/dt , and $\ln$ is the natural logarithm. The scale-free ansatz $C(t) = 3 \log_2(t/t_*) + C_*$, with arbitrary reference time $t_* > 0$ and $C_* = C(t_*)$, then gives $H(t) = 1/t$ after identifying a_{rel} with the physical cosmological scale factor. From the benchmark present age $t_0 = 13.80 \text{ Gyr}$ and defining $H_0 \equiv H(t_0)$,

$$H_0^{\text{age}} = 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (3)$$

As emphasized in the preceding note, the age is itself inferred within a cosmological model, so this is a scale connection rather than an independent measurement of H_0 . The match remains suggestive. Its strongest reading is not that three fixed channels describe every epoch—Dark Energy Spectroscopic Instrument (DESI) Data Release 2 distances already exclude that constant-power history [1]—but that cosmic expansion and the rate at which histories become distinguishable may share a natural scale.

Equation (2) makes the next requirement exact. A cosmological claim needs a $C(t)$ defined without using $H(t)$, together with a measure of compatibility that survives changes of coordinates, resolution, and time slicing. Otherwise a chosen radius or discretization can set the apparent bit rate. This note asks whether the relational idea can be carried from a combinatorial neighborhood into classical dynamics in a principled way.

CONCEPTUAL PERSUASIONS

- The original clue remains $H_0 t_0 \simeq 1$: the present expansion rate is naturally comparable to the inverse accumulated

age.

- ▶ A clue becomes a physical model only when “compatible history” has an independent operational meaning.
- ▶ The decisive object is not merely a count. It is a count or measure that does not change when the same physics is described with different coordinates or numerical resolution.
- ▶ This shifts the problem from inventing a better cosmological fit to identifying the correct measure on histories.

Conditional survival of histories

Represent history h_i by states $x_i(s)$ at ordered slices $s = 0, 1, \dots$, where a slice is one step in the chosen temporal coarse-graining. Consider pairs h_i, h_j drawn from an ensemble conditioned to share the same resolved past. Let F_s be a predictor fixed before the comparison that maps a state at slice s to its predicted successor. Define the innovation—the part not predicted from the preceding resolved state—by

$$\epsilon_i(s+1) = x_i(s+1) - F_s[x_i(s)]. \quad (4)$$

For a nonnegative dissimilarity $d(u, v) \geq 0$ between two innovations, the cumulative conditional distance through discrete slice N is

$$\delta_N(i, j) = \sum_{s < N} d(\epsilon_i(s+1), \epsilon_j(s+1)). \quad (5)$$

At fixed tolerance $r \geq 0$, the survival fraction and information are

$$\rho_N(r) = \Pr[\delta_N(i, j) \leq r \mid \text{shared resolved past}], \quad C_N(r) = -\log_2 \rho_N(r). \quad (6)$$

where $\Pr$ is probability over the conditioned pair ensemble. Because every summand in Eq. (5) is nonnegative, the compatible sets are nested in time. The construction is therefore the history analogue of the original neighborhood filtration.

Four laptop-scale controls, using fixed seed 20260824 and ensembles of up to two million history pairs, tested what the estimator counts. Predictable uniform expansion raised raw fixed-radius complexity from 1.0000 to 1.9546 bits, but exact conditioning reduced it to zero. A deterministic chaotic standard map—an area-preserving discrete dynamical system—produced 0.5581 raw bits at the test radius and again zero after conditioning. Independent binary innovations injected at exactly one bit per update were recovered at 1.0022 bits per update; grouping two updates at a time returned 1.0010 bits per primitive update. Finally, a deterministic hidden bit tape yielded 0.9971 bits per update to an observer conditioned only on the revealed macro-history, while an observer given the complete microstate measured zero.

Table 1: Controlled tests of conditional history survival. Raw separation can be generated by known expansion or deterministic chaos; conditional novelty removes both and recovers genuinely unresolved distinctions.

System	Unconditioned or coarse result	Correctly conditioned result
Predictable uniform expansion	1.9546 raw bits	0 bits
Deterministic standard map	0.5581 raw bits	0 bits
One injected random bit/update	—	1.0022 bits/update
Fixed hidden bit tape	0.9971 bits/update	0 with full microstate

The distinction is important. Histories can separate because an already known flow carries them apart, because deterministic chaos amplifies an initial difference, or because information unavailable in their shared past is newly revealed. Only the last contribution belongs in a conditional history rate. The estimator passes this conceptual test.

CONCEPTUAL PERSUASIONS

- ▶ “Histories become different” is not yet an information statement. The difference must be evaluated relative to what their shared past and known dynamics already predict.
- ▶ Expansion and chaos are capable of producing large raw separation with zero conditional novelty.
- ▶ Hidden deterministic structure can still be new information to a coarse-grained observer. Information is conditional on the accessible past, not synonymous with microscopic randomness.
- ▶ The survival estimator measures the intended distinction in controlled systems. Its remaining issue is how compatibility itself is fixed.

Why a geometric radius is not enough

Exact conditioning is an idealization. Repeating the experiment with a slightly incorrect Hubble predictor quantified the leakage. At a central normalized radius $r = 0.005$, a 0.5% rate error generated 0.0880 bits/Gyr of false novelty; a 1% error generated 0.3117 bits/Gyr. The latter already exceeds the approximate late-time target of 0.25 bits/Gyr. This target is a reconstruction, not a prediction: using $H_\Lambda = H_0 \sqrt{\Omega_\Lambda}$ for the asymptotic constant component of a flat cosmology, where Ω_Λ is the present fractional dark-energy density, the bridge gives $\dot{C}_\Lambda = 3H_\Lambda / \ln 2 \simeq 0.25$ bits/Gyr. Across radii 0.0025, 0.005, and 0.010, that target was first crossed by Hubble-rate errors of 0.5%, 1%, and 2%, respectively.

The sharper test removed H from the predictor entirely: it set F_s to the identity map, $F_s(x) = x$. The reference rate was $H_{\text{ref}} = 1/(13.8 \text{ Gyr})$. The experiment scanned 151 fixed radii over $0.01 \leq r \leq 5$ in units of the root-mean-square initial pair separation, $[\mathbb{E}\|x_i(0) - x_j(0)\|^2]^{1/2}$, where $\mathbb{E}$ denotes an ensemble average and $\|\cdot\|$ the Euclidean norm. A nontrivial rate was required to remain within 10% across one decade of radius. No expansion history passed. Each history could nevertheless be made to return approximately 0.25 bits/Gyr by selecting a different radius (Table 2).

Table 2: Multiscale identifiability test. The radius that reproduces the same target rate changes systematically with the imposed expansion. One radius calibrated at H_{ref} does not transfer linearly.

Expansion	Radius giving $\simeq 0.25$ (RMS units)	Rate at $r = 0.5799$ (bits/Gyr)	Linear target (bits/Gyr)
$0.5H_{\text{ref}}$	0.2429	0.0237	0.1250
$1.0H_{\text{ref}}$	0.5799	0.2537	0.2500
$1.5H_{\text{ref}}$	1.0795	0.4921	0.3750

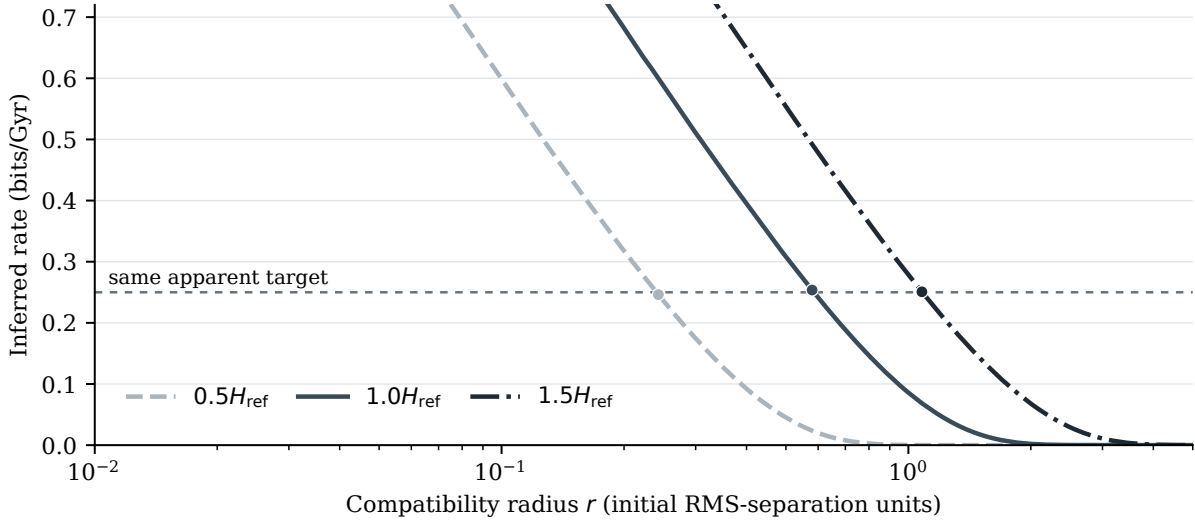


Figure 1: The inferred rate is a continuous function of the geometric tolerance. Each expansion history crosses the same apparent target (dots) at a different radius, and none develops a stable one-decade plateau. The radius therefore selects the reported rate rather than revealing an invariant one.

Initial-resolution normalization collapsed exactly, and changing the slicing from 20 to 80 steps altered cumulative distance by only 1.49×10^{-14} . The observed radius dependence is therefore the mathematical behavior of the continuous-distance survival probability, not a numerical artifact. Geometric tolerance is useful once a physical resolution is already known; it cannot by itself supply that resolution.

This conclusion points naturally to conservation laws. Rather than declaring histories compatible because they lie within an adjustable distance, one can ask whether they satisfy the constraints that make them dynamically admissible.

CONCEPTUAL PERSUASIONS

- A percent-level error in an assumed expansion law can manufacture the entire information rate the model seeks to explain.

- Scanning the full radius curve is decisive: there is no stable plateau hidden behind a fortunate choice of tolerance.
- The desired rate needs a physically specified admissible set, not a geometric radius chosen after seeing the result.
- Conservation provides exactly that kind of specification. It says which transitions are allowed before their cardinality is counted.

Noether symmetry defines admissible transitions

Let x be position and p its conjugate momentum, so (x, p) is one canonical pair in phase space, the space of instantaneous mechanical states. For a time-independent Hamiltonian $H(x, p)$, the function that assigns energy and generates time evolution, time-translation symmetry implies conservation of energy by Noether’s theorem [2, 3]. In a bounded discretized phase space $\mathcal{S}_\epsilon$, the companion note on energy-transition cardinality defines the conserving candidates by

$$\mathcal{C}_{\epsilon, \tau}(E_0) = \{w \in \mathcal{S}_\epsilon : |H(w) - E_0| \leq \tau\}, \quad \rho_{\epsilon, \tau} = \frac{|\mathcal{C}_{\epsilon, \tau}|}{|\mathcal{S}_\epsilon|}. \quad (7)$$

Here ϵ is the linear cell scale, E_0 is the conserved energy, $\tau \geq 0$ is an energy half-width, and $|\cdot|$ denotes the number of discrete states; w denotes one such discretized state. Its survival information is $C_{\epsilon, \tau} = -\log_2 \rho_{\epsilon, \tau}$, shortened to C_ϵ when admissibility is fixed by exact cell intersection. The conservation law acts as a dimensional filter: one independent scalar equation reduces the dimension by one, so admissible states concentrate on a codimension-one energy shell [4].

Here the tolerance is removed as a free parameter. Let each grid point denote a full phase-space cell of width ϵ . A cell is admissible precisely when the range of H over that cell intersects the exact energy E_0 . The effective energy uncertainty is therefore determined by discretization geometry. For the unit harmonic oscillator,

$$H(x, p) = \frac{x^2 + p^2}{2}, \quad E_0 = 1, \quad (8)$$

the minimum and maximum of H on every rectangular cell can be evaluated exactly. The calculation uses the bounded domain $\mathcal{S} = [-2, 2] \times [-2, 2]$ and units in which mass and oscillator frequency are one.

The count confirms the expected structure. Refining ϵ from 0.2 to 0.0125 raises the raw information from 2.6666 to 6.8125 bits. At the three finest resolutions, every halving adds approximately one bit. In general, for a smooth single-constraint shell in an n -dimensional domain,

$$|\mathcal{S}_\epsilon| \sim V \epsilon^{-n}, \quad |\mathcal{C}_\epsilon| \sim A_{\text{grid}} \epsilon^{-(n-1)}, \quad C_\epsilon = \log_2 \frac{1}{\epsilon} - \log_2 \frac{A_{\text{grid}}}{V} + o(1). \quad (9)$$

Here n is the phase-space dimension, V is the coordinate volume of the bounded domain, A_{grid} is the shell-content coefficient induced by the chosen grid, $\sim$ denotes asymptotic equality as $\epsilon \rightarrow 0$, and $o(1)$ is a term that vanishes in that limit. The coefficient of $\log_2(1/\epsilon)$ is not a nuisance discovered by the simulation; it is the signature of one independent conservation constraint.

Table 3: Energy-shell cardinality for the harmonic oscillator. The raw count acquires one bit per spatial refinement; subtracting the universal codimension term leaves a convergent grid-dependent finite part.

ϵ	Conserving/total cells	C_ϵ (bits)	$C_\epsilon - \log_2(1/\epsilon)$
0.2000	63/400	2.6666	0.3446
0.1000	120/1600	3.7370	0.4150
0.0500	224/6400	4.8365	0.5146
0.0250	448/25600	5.8365	0.5146
0.0125	911/102400	6.8125	0.4906

CONCEPTUAL PERSUASIONS

- Noether’s theorem supplies more than a metaphor for compatibility. Time symmetry selects an energy shell before any counting begins.
- Grid cells that intersect the shell define “almost conserving” without an adjustable energy tolerance: the uncertainty is fixed by the cell.
- One bit per halving is the numerical image of codimension one. The growth has a geometric explanation rather than

a fitted rate.

- The finite remainder is the next candidate—but it must still pass the invariance tests required of a physical quantity.

From one transition to stationary-action histories

Energy-shell cardinality concerns candidate states. Accumulated history asks which sequences of states remain dynamically admissible. For the unit harmonic-oscillator Lagrangian—the function whose time integral is the action $S = \int L dt$ —

$$L(x, \dot{x}) = \frac{1}{2}\dot{x}^2 - \frac{1}{2}x^2, \quad (10)$$

where $\dot{x} = dx/dt$, stationarity ($\delta S = 0$ under small path variations with fixed endpoints) of the discretized action gives the central Euler–Lagrange residual

$$R_m = x_{m+1} + (-2 + \Delta t^2)x_m + x_{m-1} = 0. \quad (11)$$

Here m labels a time slice, Δt is the constant interval between slices, $x_m = x(m\Delta t)$, and R_m measures failure of the discrete equation of motion. Given three position cells of width ϵ , interval arithmetic determines the exact range of R_m . A candidate continuation is admitted when that range contains zero. The associated residual half-width,

$$\tau_R(\epsilon, \Delta t) = \frac{\epsilon}{2} [2 + |-2 + \Delta t^2|], \quad (12)$$

is derived from the cells and Eq. (11); it is not adjusted to the result.

Starting from cells containing $x(0) = 1$ and $x(\Delta t) = \cos \Delta t$ in the position domain $[-1.5, 1.5]$, a dynamic program propagates the fraction of next cells, proposed uniformly from that domain, that satisfy every accumulated residual constraint. At $\Delta t = 0.2$, raw information per accepted update grows from 2.0919 to 5.9069 bits over the tested refinement range. Subtracting the same $\log_2(1/\epsilon)$ term produces a stable finest-grid value near -0.415 bits per update.

Time refinement reveals a second continuum factor. At fixed duration 2.4 and $\epsilon = 0.025$, halving Δt approximately doubles the number of constraints and therefore the accumulated information per unit time (Table 4). A continuum path is constrained at infinitely many slices, so raw path cardinality requires a temporal measure just as state cardinality requires a phase-space measure.

Table 4: Fixed-duration stationary-action histories. Spatial admissibility is physically defined, but raw information per unit time retains the normalization of the temporal slicing.

Δt	Accepted updates	Total information (bits)	Bits/unit time
0.400	5	24.6457	10.2690
0.200	11	54.0137	22.5057
0.100	23	113.5199	47.3000

This is significant progress in the definition of the problem. The arbitrary radius has disappeared; what remains is the familiar continuum question of how a measure is normalized. The spatial divergence exposes constraint codimension. The temporal divergence exposes the density of imposed slices. Both can be separated from finite dynamical content, but only an invariant construction can decide which finite content is physical.

CONCEPTUAL PERSUASIONS

- Stationary action turns a one-step conservation shell into a filtration of entire admissible paths.
- The interval tolerance is calculated from the discretized equation of motion. There is no radius left to tune to a desired bit rate.
- Spatial and temporal divergences are structurally informative: they count constraint codimension and constraint density.
- A continuum history theory therefore needs a measure on paths, not a literal count of coordinate cells or time slices.

Canonical invariance selects Liouville measure

The convergent finite cell-count term is tempting, but convergence within one grid is weaker than physical invariance. The same harmonic shell was therefore counted with equal-area rectangular cells of aspect ratios 0.5, 1, and 2, and after the canonical shear

$$Q = x, \quad P = p + x, \quad dx \wedge dp = dQ \wedge dP. \quad (13)$$

Here (Q, P) are new canonical coordinates and $\wedge$ is the antisymmetric differential product defining oriented phase area; the last equality states that the transformation preserves that area. At the finest common cell area, the codimension-subtracted finite terms were 0.1773, 0.4906, 0.1773, and 0.2225 bits, respectively. Their 0.3134-bit spread persists in the refinement limit. Intersecting-cell count measures an anisotropic grid perimeter; it is not a canonical scalar.

Hamiltonian mechanics supplies the appropriate replacement. Liouville measure

$$d\Gamma = dx dp \quad (14)$$

is the phase-area element for one canonical pair; in n pairs it is the product $d\Gamma = \prod_{k=1}^n dx_k dp_k$. It is preserved by Hamiltonian flow and by canonical transformations [3, 5]. For a physical energy half-width τ_E , define the band density

$$g_{\tau_E}(E) = \frac{1}{2\tau_E} \int \mathbf{1}\{|H(x, p) - E| \leq \tau_E\} d\Gamma. \quad (15)$$

Here $\mathbf{1}\{\cdot\}$ is one when its condition holds and zero otherwise, and $\tau_E > 0$ is a fixed physical energy half-width. Thus g_{τ_E} is phase volume per unit energy. As $\tau_E \rightarrow 0$, this approaches the microcanonical density of states

$$g(E) = \int \delta(H - E) d\Gamma, \quad (16)$$

where δ is the Dirac delta distribution, which restricts the integral to the energy surface. Unlike cell incidence, Eq. (15) weights every cell by its phase volume.

The invariance test used $\tau_E = 0.05$. For the harmonic oscillator, the exact band density is $2\pi = 6.283185$. Across all aspect ratios and the canonical shear, numerical estimates ranged from 6.283000 to 6.290000: a maximum relative error of 0.108% and cross-scheme spread of 0.111%. A second, anharmonic Hamiltonian,

$$H_{\text{anh}}(x, p) = \frac{p^2}{2} + \frac{x^2}{2} + \frac{x^4}{8}, \quad (17)$$

has reference band density 5.061875, obtained by high-accuracy quadrature of the enclosed phase volume. The same schemes returned 5.058000 to 5.071000, with maximum error 0.180% and cross-scheme spread 0.257%.

Table 5: Canonical and grid invariance at the finest tested quadrature. Raw cell finite parts vary materially; Liouville band densities agree with the invariant reference values for two Hamiltonians.

Scheme	Cell finite part	Harmonic g_{τ_E}	Anharmonic g_{τ_E}
Aspect 0.5	0.1773	6.284000	5.065000
Aspect 1	0.4906	6.290000	5.071000
Aspect 2	0.1773	6.284000	5.058000
Canonical shear	0.2225	6.283000	5.058000
Reference invariant	—	6.283185	5.061875

The split is unambiguous. Raw cardinality knows about the coordinate lattice. Liouville density knows about the Hamiltonian system. The first is a useful regularization diagnostic; the second is an invariant candidate for physical history measure.

CONCEPTUAL PERSUASIONS

- ▶ A convergent number is not automatically a physical number. It must survive coordinate transformations that leave the dynamics unchanged.
- ▶ Grid-cell incidence fails that test because different cell shapes cover the same shell differently.
- ▶ Liouville measure passes across grids, canonical coordinates, and both tested Hamiltonians.

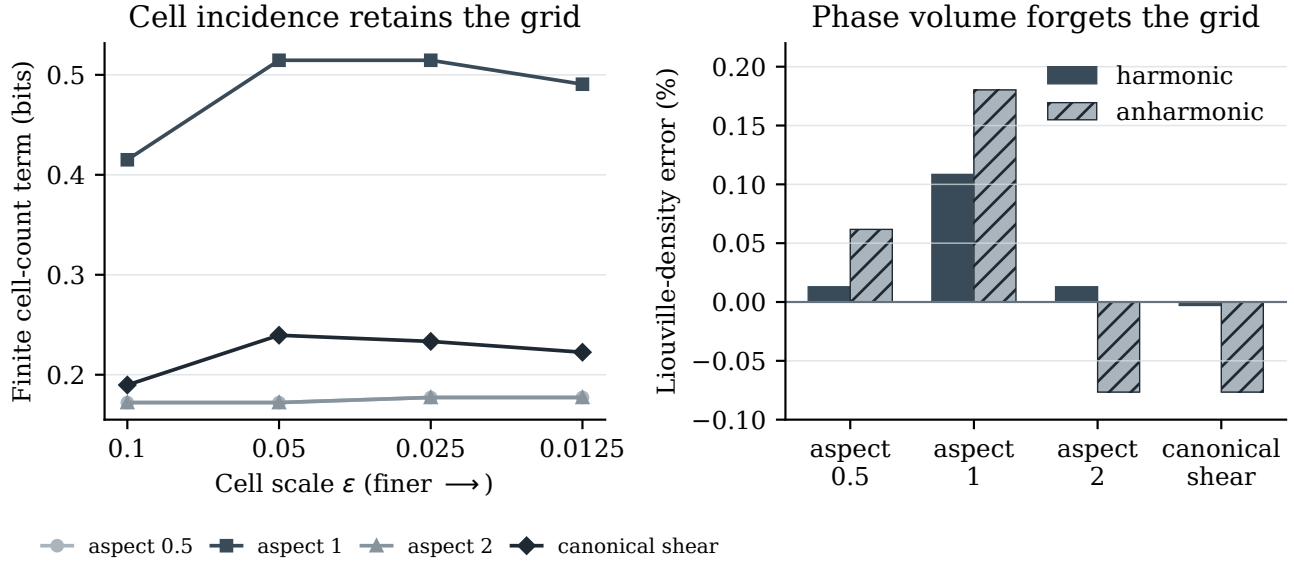


Figure 2: Two refinement tests separate regularization from physics. Left: after subtracting the universal codimension term, cell-incidence counts approach different finite values for different grids and canonical coordinates. Right: Liouville band density at the finest quadrature remains within 0.18% of the invariant reference value for every scheme and both Hamiltonians.

The revised cosmological object

The result suggests a precise refinement of Eq. (6). Let $\mathcal{P}_n$ be the set of candidate continuations at causal slice n , let $\mathcal{A}_n \subseteq \mathcal{P}_n$ be the subset satisfying the dynamical constraints given the shared past, and let μ be the invariant measure appropriate to the system. Write $\mathcal{A}_{<n}$ for all admissibility conditions imposed on slices before n . The conditional survival information added at slice n is

$$\Delta C_\mu(n) = -\log_2 \frac{\mu(\mathcal{A}_n | \mathcal{A}_{<n})}{\mu(\mathcal{P}_n | \mathcal{A}_{<n})}. \quad (18)$$

The notation $\mu(B | A)$ means the measure assigned to B after conditioning on A ; the ratio is the surviving fraction of candidate measure. For classical Hamiltonian systems, the present tests select Liouville measure over coordinate-cell count. For fields or gravity, the analogue must respect the constrained symplectic structure (the geometry defined by canonical coordinates and their conserved phase volume) and gauge redundancy (different mathematical descriptions of the same physical state). For relativistic histories it must additionally respect causal order and proper-time slicing. Through slice N , define the accumulated information by

$$C_\mu(N) = \sum_{n=1}^N \Delta C_\mu(n), \quad C_\mu(0) = 0, \quad (19)$$

and let $C_\mu(\tau)$ denote a continuum interpolation when the refinement limit exists. Thus ΔC_μ is the new conditional information at one slice, while C_μ is the accumulated history quantity.

Equation (18) is dimensionless when numerator and denominator are positive-measure sets of the same phase dimension. Exact constraint surfaces instead require an induced measure, a physical band, or a relative density such as $g(E)/g_*(E)$, where g_* is an independently specified reference density with the same units as g . An absolute number of phase-space states introduces an action cell; quantum mechanics naturally supplies a scale of order Planck's constant $\hbar$ per canonical pair, whereas a purely classical theory does not. The normalization is therefore a physical question, not a parameter to be chosen for cosmological agreement.

The cosmological bridge can now be stated with greater discipline:

$$H_{\text{rel}}(\tau) = \frac{\ln 2}{3} \frac{dC_\mu}{d\tau}, \quad (20)$$

where τ is an invariant time parameter, such as proper time, along the relevant congruence (a family of neighboring observer worldlines) and C_μ is built from conditional invariant measure. The equation remains a constitutive

hypothesis, but its information variable is no longer an arbitrary neighborhood count. Its open ingredients are sharply located: the gravitational history measure, the treatment of constraints and gauge, and the continuum normalization of successive causal slices.

The earlier Hubble-scale result gives a quantitative target for this object. At the present epoch, Eq. (20) asks for an invariant conditional rate of order 0.25–0.31 bits/Gyr, depending on whether one isolates the late-time constant component or uses the full H_0 . That number should emerge only after the measure and normalization are fixed. The new results make clear what must be derived before it is compared with cosmological data.

CONCEPTUAL PERSUASIONS

- ▶ The compatible-history idea survives in a stronger form: probability is replaced by conditional invariant measure.
- ▶ Noether constraints define admissibility; Liouville geometry defines how admissible alternatives are weighted.
- ▶ A quantum action cell is a natural candidate for making phase volume dimensionless, while a covariant path measure must resolve the temporal continuum.
- ▶ The Hubble-scale bit rate is now a target for an invariant construction, not a value used to choose its tolerance or cutoff.

Constructive outlook

The accumulated-history program began with the observation that a longer past can distinguish histories that once appeared equivalent. Its simplest cosmological realization found the correct present Hubble scale, then learned from the expansion history that the relevant information rate must be dynamical. The experiments reported here advance the next layer.

First, conditional survival separates genuinely unresolved distinctions from predictable expansion and chaos. Second, the absence of a stable geometric radius shows why compatibility must be supplied by physics. Third, Noether symmetry and stationary action supply that admissibility without fitting a tolerance. Fourth, continuum refinement separates universal constraint terms from finite content. Finally, canonical tests select Liouville measure and exclude raw cell incidence as the physical finite quantity.

The next calculation is consequently well defined. Construct the constrained symplectic measure for a finite gravitational or cosmological system, define the conditional measure added by successive causal slices, and test whether a renormalized rate converges under simultaneous phase-space and temporal refinement. Only then should the rate be compared with $H = (\ln 2/3)\dot{C}_\mu$ and the held-out DESI expansion history.

The central promise has become more concrete. Accumulated history need not be counted by arbitrary coordinates; symmetry can define the admissible roads, and invariant measure can say how much room those roads occupy. The remaining problem is no longer what should be counted. It is how the invariant measure of newly admissible history flows in a relativistic universe.

CONCEPTUAL PERSUASIONS

- ▶ The program has moved from analogy to a tested measurement hierarchy: conditional prediction, physical admissibility, continuum analysis, and canonical invariance.
- ▶ Noether’s theorem identifies the roads; Liouville measure replaces the coordinate-dependent count of paving stones.
- ▶ The surviving question is precise: does invariant compatible-history measure flow at the Hubble rate? The near equality $H_0 \simeq 1/t_0$ fixes the motivating present target.

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